Q1. What is machine learning and how is it different from traditional programming?

Answer: Machine learning is a subset of artificial intelligence where systems learn patterns from data and improve their performance without being explicitly programmed. In traditional programming, a developer writes rules manually. In machine learning, the algorithm discovers the rules automatically from examples in the data.

Q2. What are the three main types of machine learning?

Answer: The three main types are supervised learning, where the model learns from labeled data; unsupervised learning, where the model finds patterns in unlabeled data; and reinforcement learning, where an agent learns by taking actions and receiving rewards or penalties from an environment.

Q3. What is the difference between supervised and unsupervised learning?

Answer: In supervised learning, the training data includes input features along with correct output labels, and the model learns to map inputs to outputs. In unsupervised learning, there are no labels and the model finds hidden patterns or groupings in the data on its own.

Q4. What is a training set, validation set, and test set in machine learning?

Answer: The training set is used to train the model. The validation set is used during training to tune hyperparameters and prevent overfitting. The test set is held out until the very end and used to evaluate the final model performance on unseen data.

Q5. What is the difference between a classification and a regression problem?

Answer: Classification predicts a discrete category or label, such as whether an email is spam or not spam. Regression predicts a continuous numerical value, such as the price of a house. The type of output variable determines whether a problem is classification or regression.

Q6. What is overfitting in machine learning and how can it be prevented?

Answer: Overfitting occurs when a model learns the training data too well, including its noise, and performs poorly on new unseen data. It can be prevented by using more training data, applying regularization techniques, simplifying the model, using dropout in neural networks, or applying cross-validation.

Q7. What is underfitting and what causes it?

Answer: Underfitting occurs when a model is too simple to capture the underlying patterns in the data, resulting in poor performance on both training and test data. It is caused by using a model with insufficient complexity, too little training, or removing too many features from the dataset.

Q8. What is the bias-variance tradeoff in machine learning?

Answer: Bias is the error caused by overly simplistic assumptions in the model, leading to underfitting. Variance is the error caused by excessive sensitivity to training data, leading to overfitting. The tradeoff means reducing bias often increases variance and vice versa, so the goal is to find the right balance for best generalization.

Q9. What is a feature in machine learning and what is feature engineering?

Answer: A feature is an individual measurable input variable used by the model to make predictions. Feature engineering is the process of creating, selecting, or transforming raw data into better features that improve model accuracy and help the algorithm learn patterns more effectively.

Q10. What is feature selection and why is it important?

Answer: Feature selection is the process of identifying and keeping only the most relevant input variables for the model while removing irrelevant or redundant ones. It is important because it reduces model complexity, speeds up training, improves accuracy, and makes the model easier to interpret.

Q11. What is the difference between a parametric and a non-parametric model?

Answer: A parametric model assumes a fixed form for the underlying data distribution and has a set number of parameters, such as linear regression. A non-parametric model makes no assumptions about the distribution and can grow in complexity with more data, such as k-nearest neighbors or decision trees.

Q12. What is linear regression and what does the model equation represent?

Answer: Linear regression is a supervised algorithm that models the relationship between one or more input features and a continuous output by fitting a straight line. The equation $Y = b0 + b1X$ represents where b0 is the intercept and b1 is the slope, showing how much the output changes per unit increase in the input.

Q13. What is logistic regression and how is it used for classification?

Answer: Logistic regression is a supervised algorithm that predicts the probability that an input belongs to a particular class. Despite its name, it is a classification algorithm that uses the sigmoid function to squash output values between 0 and 1, then applies a threshold to assign class labels.

Q14. What is a decision tree and how does it make predictions?

Answer: A decision tree is a tree-shaped model that splits data into branches based on feature conditions at each node. It makes predictions by traversing from the root node down through the branches following the conditions that match the input, until it reaches a leaf node that gives the final output.

Q15. What is the difference between a decision tree and a random forest?

Answer: A decision tree is a single model that can overfit easily on complex data. A random forest is an ensemble of many decision trees trained on random subsets of the data and features. It combines their predictions by voting or averaging, which reduces overfitting and produces a more accurate and stable model.

Q16. What is ensemble learning and what are its two main types?

Answer: Ensemble learning combines multiple models to produce better predictions than any single model alone. The two main types are bagging, which trains models in parallel on random subsets of data to reduce variance (e.g., random forest), and boosting, which trains models sequentially where each corrects the errors of the previous one to reduce bias (e.g., XGBoost).

Q17. What is the difference between bagging and boosting?

Answer: Bagging trains multiple models independently in parallel on different random samples of data and combines their outputs to reduce variance and prevent overfitting. Boosting trains models sequentially, where each new model focuses on the errors made by the previous one, reducing bias and improving accuracy progressively.

Q18. What is the k-nearest neighbors (KNN) algorithm and how does it work?

Answer: KNN is a simple non-parametric algorithm that classifies or predicts a data point based on the majority class or average value of its k closest neighbors in the feature space. It uses a distance metric like Euclidean distance to find neighbors and makes no assumptions about the underlying data distribution.

Q19. What is the support vector machine (SVM) and what is a hyperplane?

Answer: SVM is a supervised classification algorithm that finds the optimal hyperplane that best separates two classes in the feature space with the maximum margin. A hyperplane is a decision boundary that divides the feature space into class regions, and SVM maximizes the distance between the hyperplane and the nearest data points of each class called support vectors.

Q20. What is the kernel trick in SVM and why is it used?

Answer: The kernel trick is a method that transforms data into a higher-dimensional space using a kernel function so that a linear hyperplane can separate classes that are not linearly separable in the original space. It allows SVM to solve complex non-linear classification problems without explicitly computing the high-dimensional transformation.

Q21. What is regularization in machine learning and what are L1 and L2 regularization?

Answer: Regularization adds a penalty to the model's loss function to discourage overly complex models and reduce overfitting. L1 regularization (Lasso) adds the sum of absolute values of coefficients and can shrink some to exactly zero, performing feature selection. L2 regularization (Ridge) adds the sum of squared coefficients and shrinks all coefficients toward zero without eliminating them.

Q22. What is cross-validation and what is k-fold cross-validation?

Answer: Cross-validation is a technique to evaluate a model's performance on unseen data by splitting the dataset into multiple parts. In k-fold cross-validation, the data is divided into k equal folds, the model is trained on k-1 folds and tested on the remaining fold, and this process repeats k times so every fold serves as the test set once.

Q23. What is a confusion matrix and what are the four values it contains?

Answer: A confusion matrix is a table used to evaluate the performance of a classification model. It contains four values: true positives (correctly predicted positive cases), true negatives (correctly predicted negative cases), false positives (negative cases incorrectly predicted as positive), and false negatives (positive cases incorrectly predicted as negative).

Q24. What is the difference between precision and recall?

Answer: Precision is the proportion of predicted positive cases that are actually positive, measuring how many of the model's positive predictions are correct. Recall is the proportion of actual positive cases that the model correctly identified, measuring how well the model finds all positive cases. Precision focuses on prediction quality; recall focuses on coverage.

Q25. What is the F1 score and when is it preferred over accuracy?

Answer: The F1 score is the harmonic mean of precision and recall, balancing both metrics into a single number. It is preferred over accuracy when the dataset is imbalanced, meaning one class has far more examples than the other, because accuracy can be misleadingly high by simply predicting the majority class every time.

Q26. What is the ROC curve and what does the AUC value represent?

Answer: The ROC (Receiver Operating Characteristic) curve is a plot of the true positive rate against the false positive rate at various classification thresholds. The AUC (Area Under the Curve) represents the overall ability of the model to distinguish between classes, where a value of 1 means perfect classification and 0.5 means no better than random guessing.

Q27. What is clustering and what is the difference between k-means and hierarchical clustering?

Answer: Clustering is an unsupervised technique that groups similar data points together without using labels. K-means partitions data into a fixed number of k clusters by iteratively assigning points to the nearest centroid and updating centroids. Hierarchical clustering builds a tree of clusters by either merging smaller clusters bottom-up or splitting larger ones top-down, without needing to specify k in advance.

Q28. What is dimensionality reduction and what is PCA?

Answer: Dimensionality reduction is the process of reducing the number of input features while retaining as much information as possible. Principal Component Analysis (PCA) is the most common technique, transforming original features into a smaller set of uncorrelated components called principal components that capture the maximum variance in the data.

Q29. What is hyperparameter tuning and what are common methods used?

Answer: Hyperparameter tuning is the process of finding the best configuration settings for a machine learning model that are not learned from data but set before training, such as learning rate or number of trees. Common methods include grid search, which tries all combinations of specified values; random search, which samples random combinations; and Bayesian optimization, which uses past results to intelligently choose the next combination to try.

Q30. What is the difference between a model parameter and a hyperparameter?

Answer: A model parameter is a value learned automatically from the training data during the model fitting process, such as the weights in linear regression or a neural network. A hyperparameter is a setting defined by the analyst before training begins that controls how the model learns, such as the number of layers, learning rate, or regularization strength.